

# Daily Review



- 1 `delay(1000)` waits...
- 2 `setup()` runs how often?
- 3 `loop()` runs...
- 4 `digitalWrite(4, HIGH);` does...
- 5 The exam's pause 500 is...

1 s

Once

For ever

Switches pin 4 on

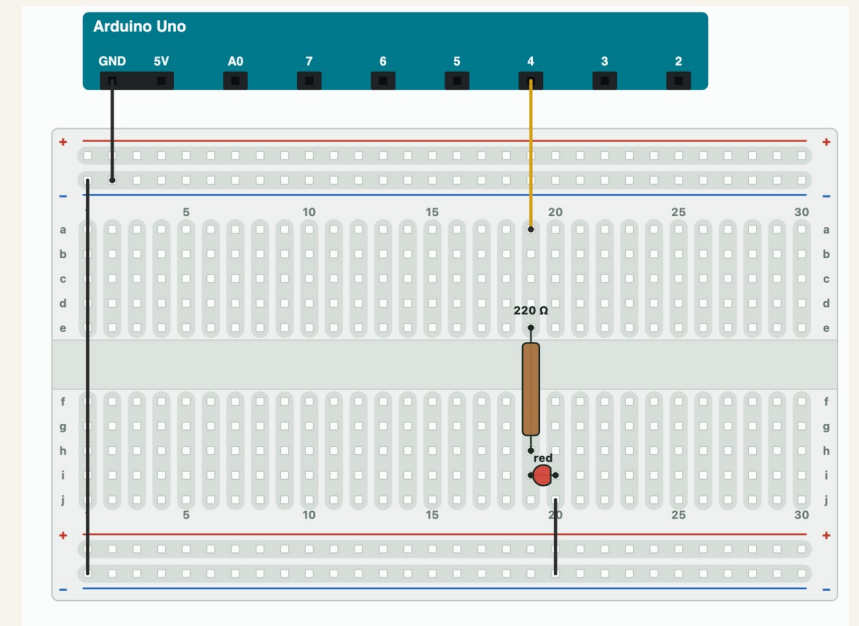
0.5 s

# Build, Test and Fix

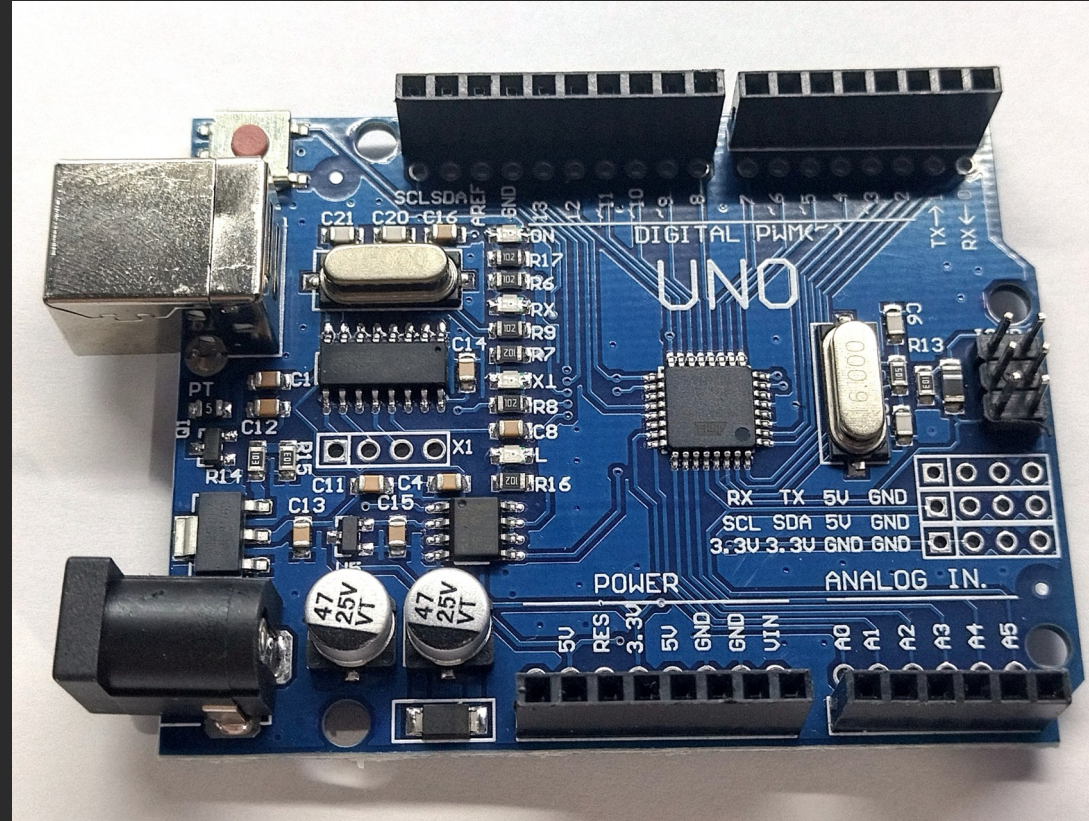
---

Put the program on a real Uno. Then make it work.

National 5 Engineering Science



This board has no program on it yet. What must you give it to flash an LED?



An Arduino Uno

## PART 1

# Build It

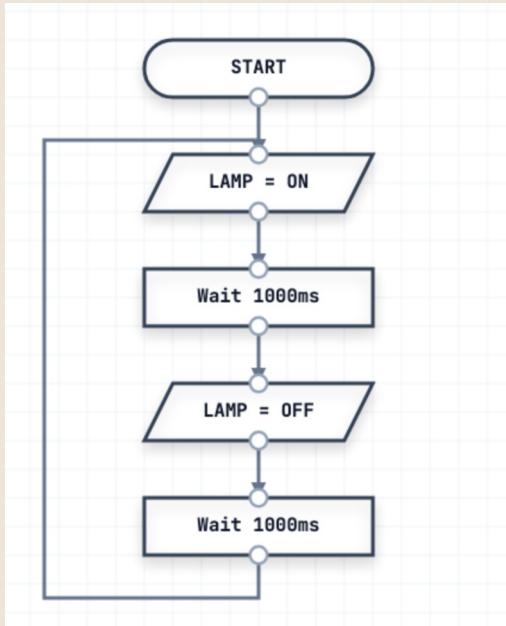
Flowchart, code, wires, upload.

## By the end of this part...



- 1 I can wire an LED to a pin on the Uno
- 2 I can match each line of code to a flowchart box
- 3 I can upload a program and check it works

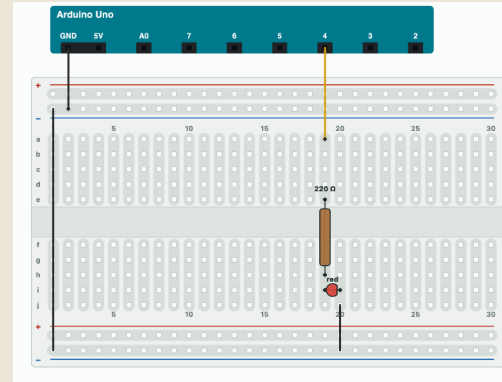
# From Flowchart to Board



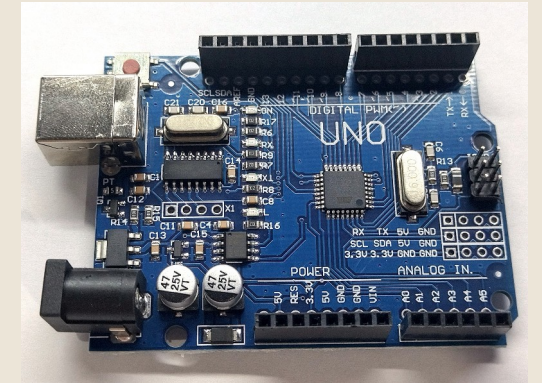
1 Flowchart

```
setup()  
loop()  
delay(1000);
```

2 Code



3 Wires

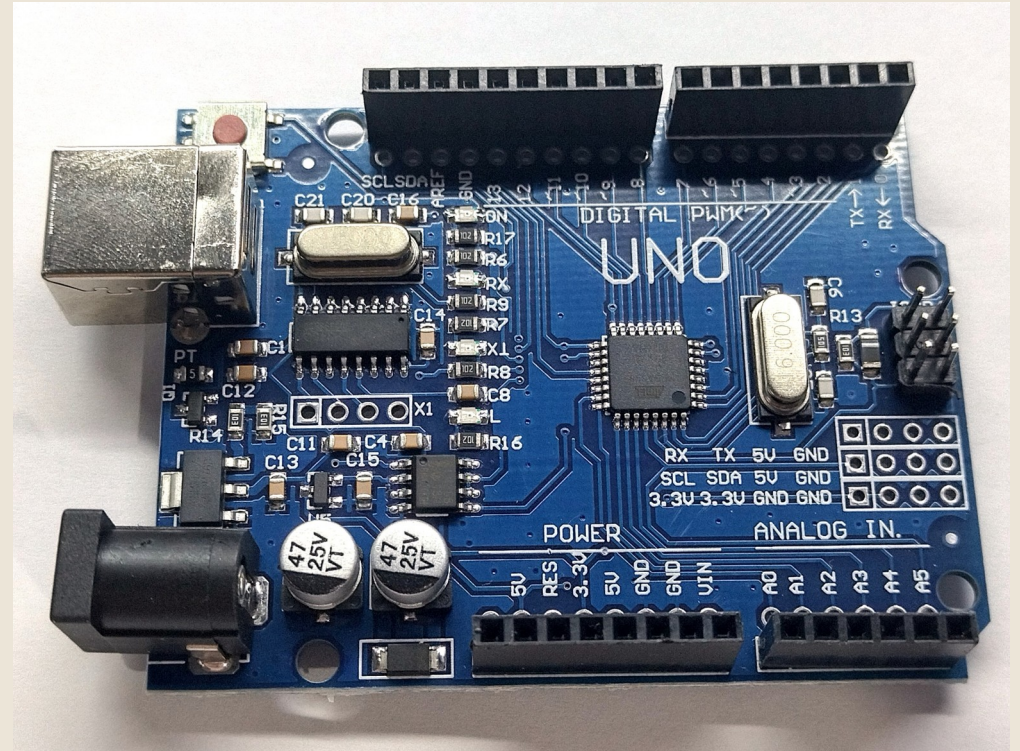


4 Upload

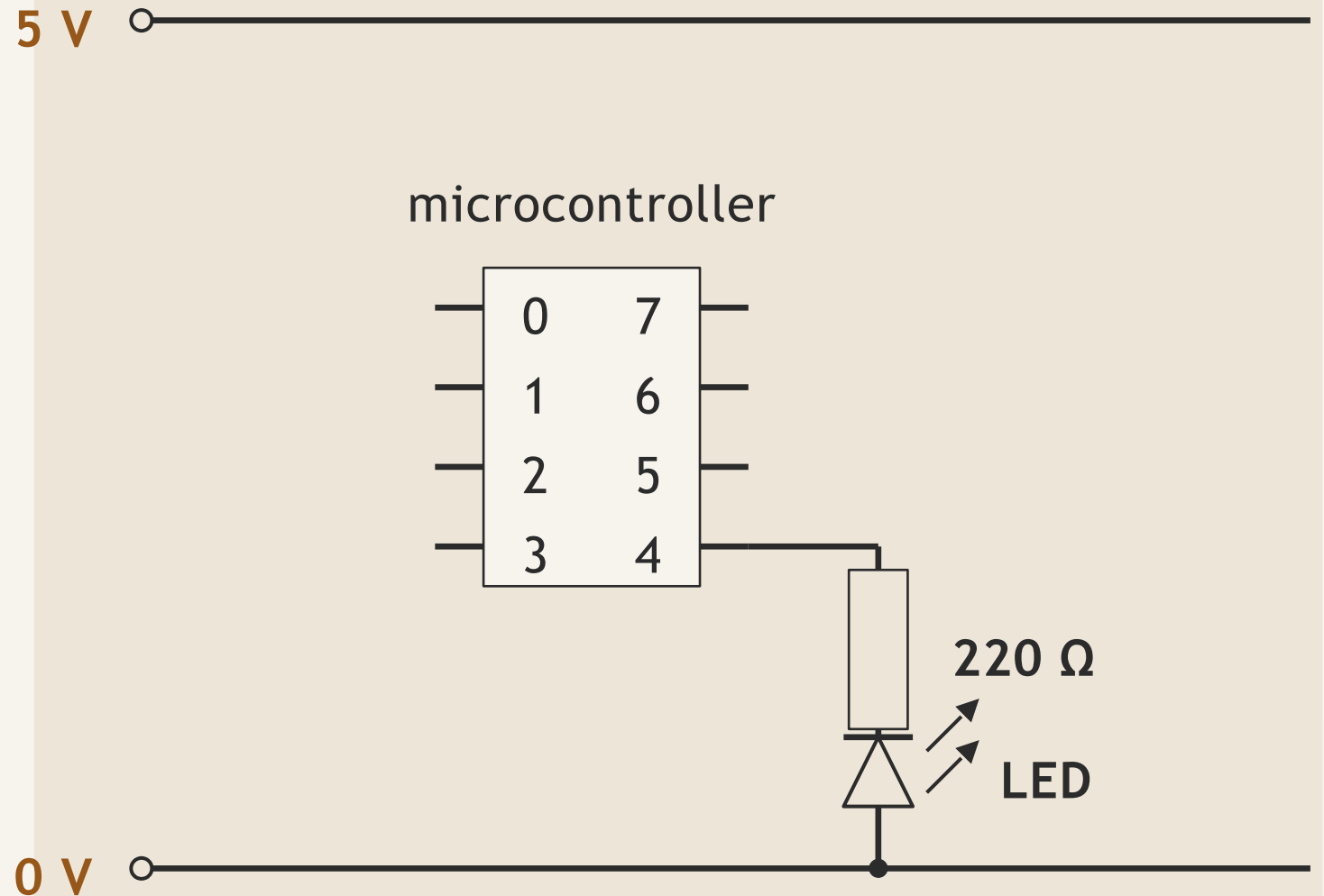
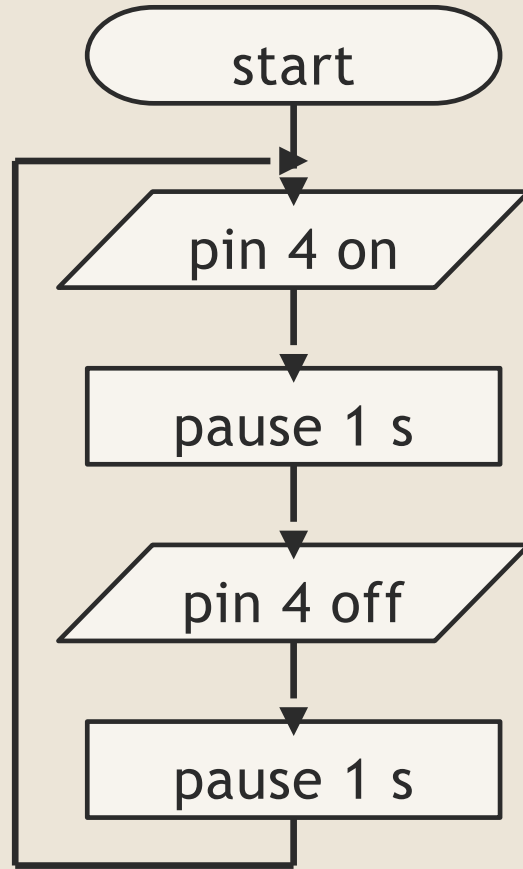
# What to Collect

Collect one set per pupil.

- 1 Uno and its USB lead
- 2 Breadboard
- 3 3 LEDs: red, amber, green
- 4 3 × 220  $\Omega$  resistors
- 5 6 jumper leads

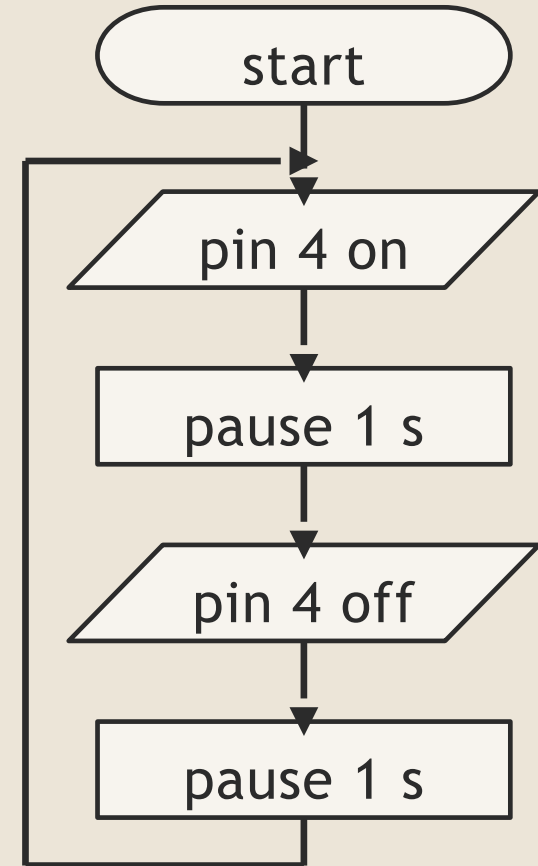


# Worked Example 1: The Plan and the Wires

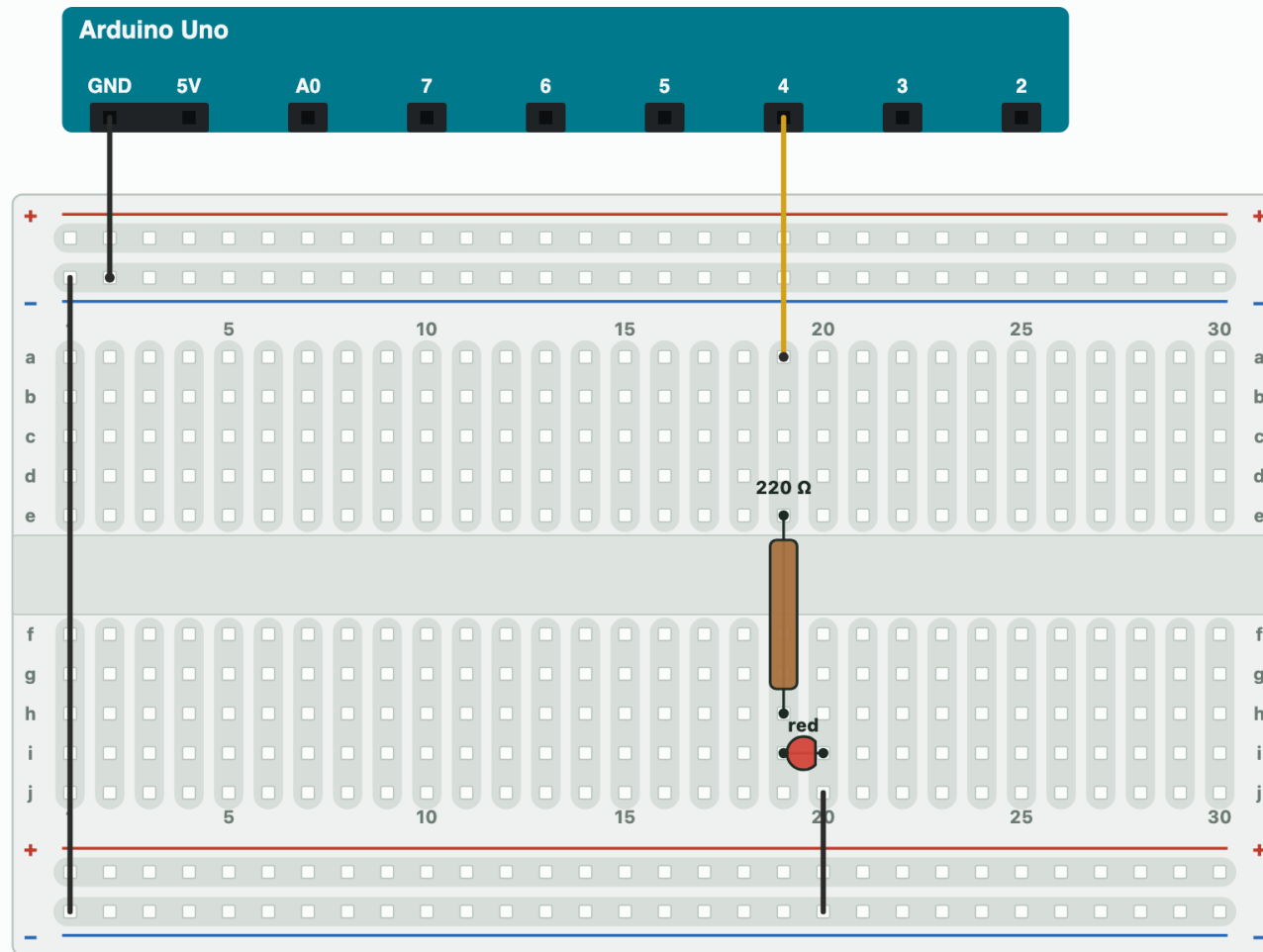


## Worked Example 1: The Code

```
1 void setup() {  
2   pinMode(4, OUTPUT);  
3 }  
4 void loop() {  
5   digitalWrite(4, HIGH);  
6   delay(1000);  
7   digitalWrite(4, LOW);  
8   delay(1000);  
9 }
```



# Worked Example 1: Build It



**Yellow lead**

pin 4 to the 220  $\Omega$

**LED**

long leg to the resistor

**Black lead**

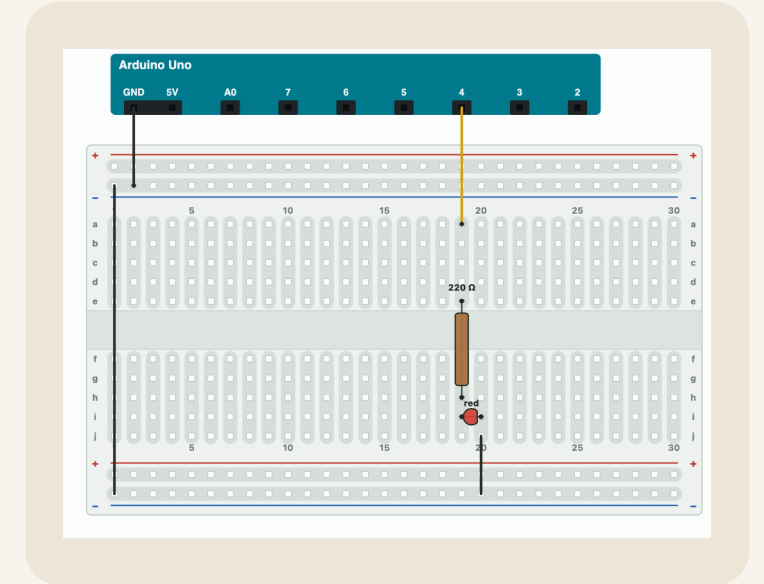
GND to the - rail

# Build It: Worked Example 1



## Practical sheet 1

Wire it. Type the code (page 15).  
Upload. 1 s on, 1 s off?



**SAFETY** USB power only. Unplug before you move a wire.

15 min



# Try Changing This: Worked Example 1



1 Both delays 200. Faster or slower?

Faster: 5 flashes a second

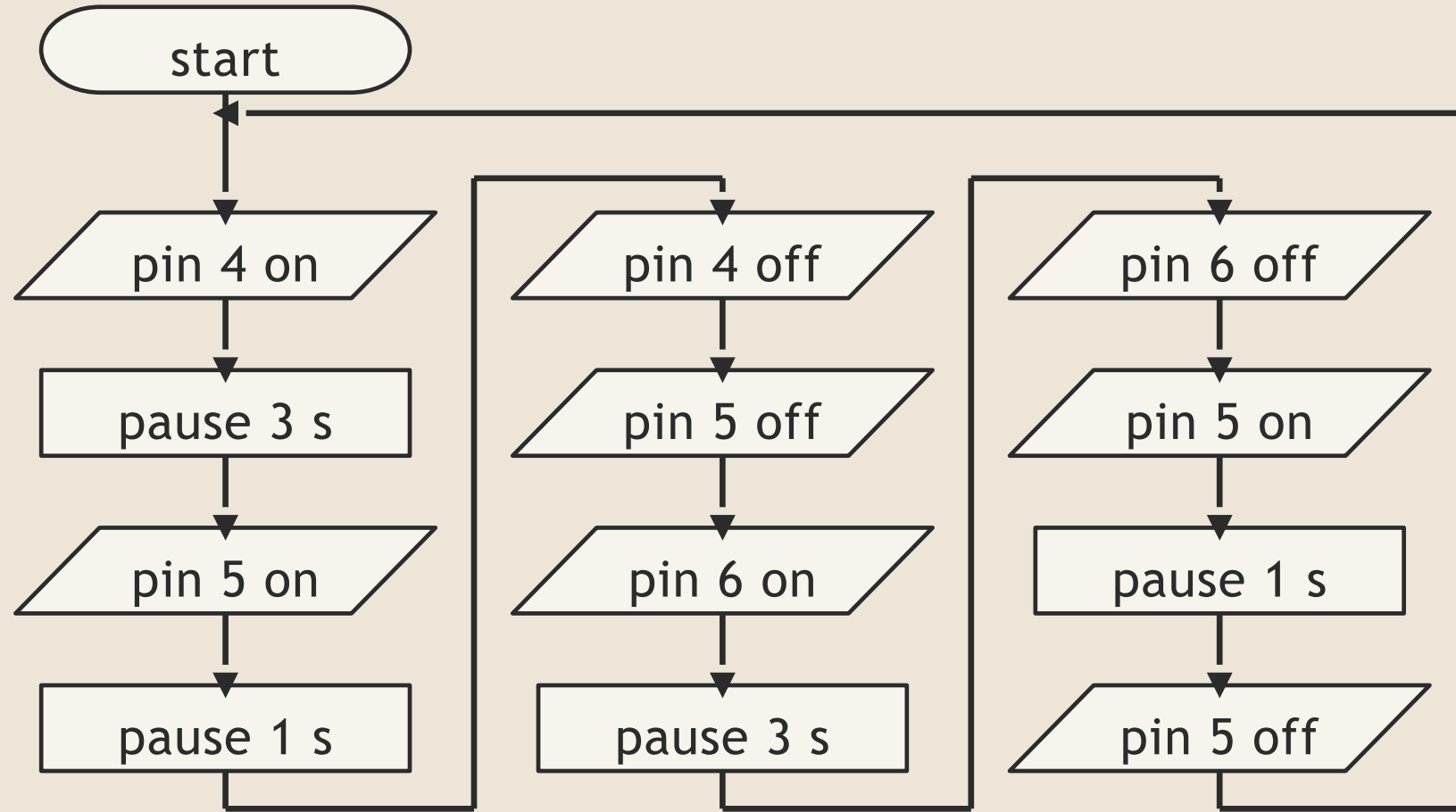
2 `delay(100)`, then `delay(900)`?

A short blink once a second

3 Move the LED to pin 5. What else changes?

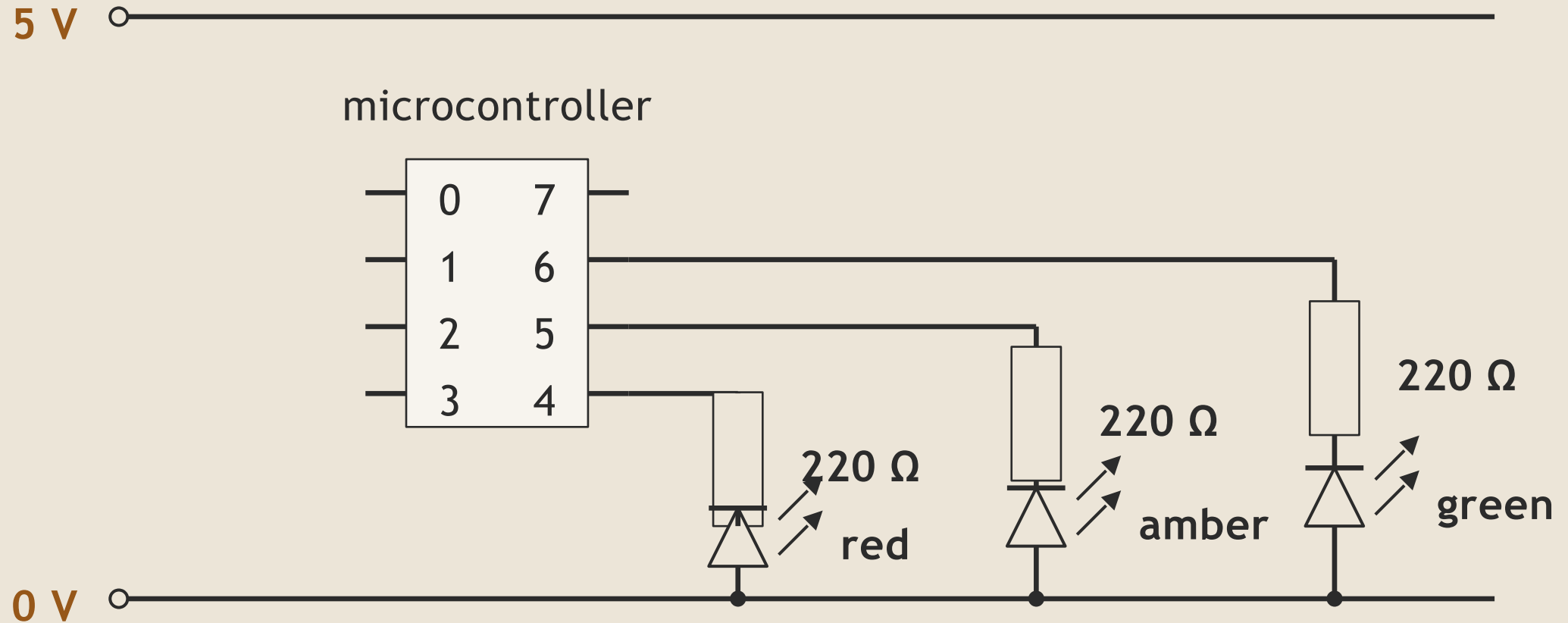
All three pin 4s — code must match the wire

## Worked Example 2: Traffic Lights



- 1 red
- 2 red +  
amber
- 3 green
- 4 amber

## Worked Example 2: The Wires



```
pinMode(4, OUTPUT); pinMode(5, OUTPUT); pinMode(6, OUTPUT);
```

## Worked Example 2: The Code

```
1 void loop() {  
2   digitalWrite(4, HIGH);  
3   delay(3000);  
4   digitalWrite(5, HIGH);  
5   delay(1000);  
6   digitalWrite(4, LOW);  
7   digitalWrite(5, LOW);
```

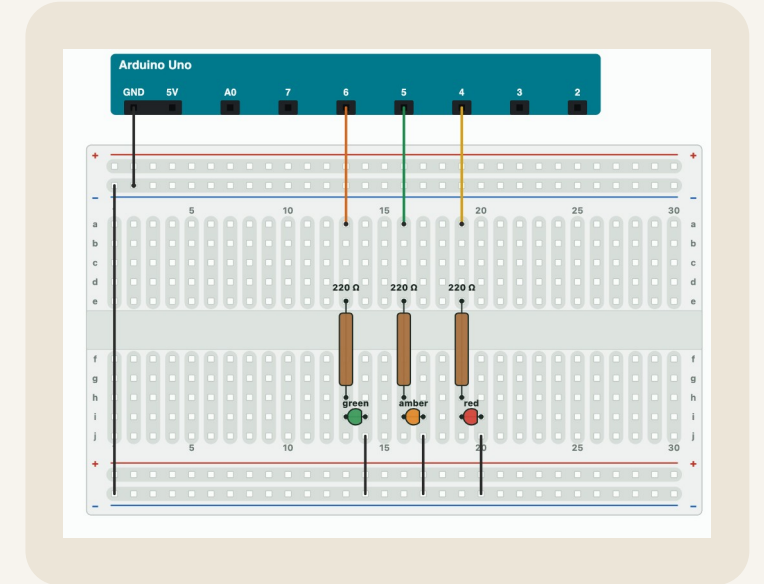
```
8   digitalWrite(6, HIGH);  
9   delay(3000);  
10  digitalWrite(6, LOW);  
11  digitalWrite(5, HIGH);  
12  delay(1000);  
13  digitalWrite(5, LOW);  
14 }
```

# Build It: Worked Example 2



## Practical sheet 1

Add amber and green (page 16).  
Upload. Is the order right?



**SAFETY** USB power only. Unplug before you move a wire.

15 min



## Try Changing This: Worked Example 2



1

Green twice as long as red. Which line?

Line 9: `delay(6000)`

2

Add `//` by your name. Does it still run?

Yes — `//` is a comment

3

Swap the 4 and 6 in `setup()` only?

Nothing — all three are still outputs

# Checkpoint



- 1 Why the 220  $\Omega$  resistor?
- 2 A pin is not set in setup(). What happens?
- 3 Two LEDs on at once in a flowchart?

Stops too much current  
— protects the LED

Its LED stays off

Two output boxes, one  
after the other

# Plenary: Part 1



1 `setup()` sets each pin as...

Input or output

2 The closing `}` of `loop()` acts like...

The arrow back up

3 `delay(3000)` waits...

3 s

4 The code's pin must match...

The wire

5 Verify, then...

Upload

# Look back — by the end of this part...



- 1 I can wire an LED to a pin on the Uno
- 2 I can match each line of code to a flowchart box
- 3 I can upload a program and check it works

# Daily Review



- 1 `pinMode(5, OUTPUT);` goes in...
- 2 `delay(250)` waits...
- 3 Red and amber on: how many boxes?
- 4 Why a  $220\ \Omega$  with each LED?
- 5 The code uses pin 5, the wire pin 4?

`setup()`

0.25 s

Two

To protect it

Nothing happens

PART 2

# Sensors and Faults

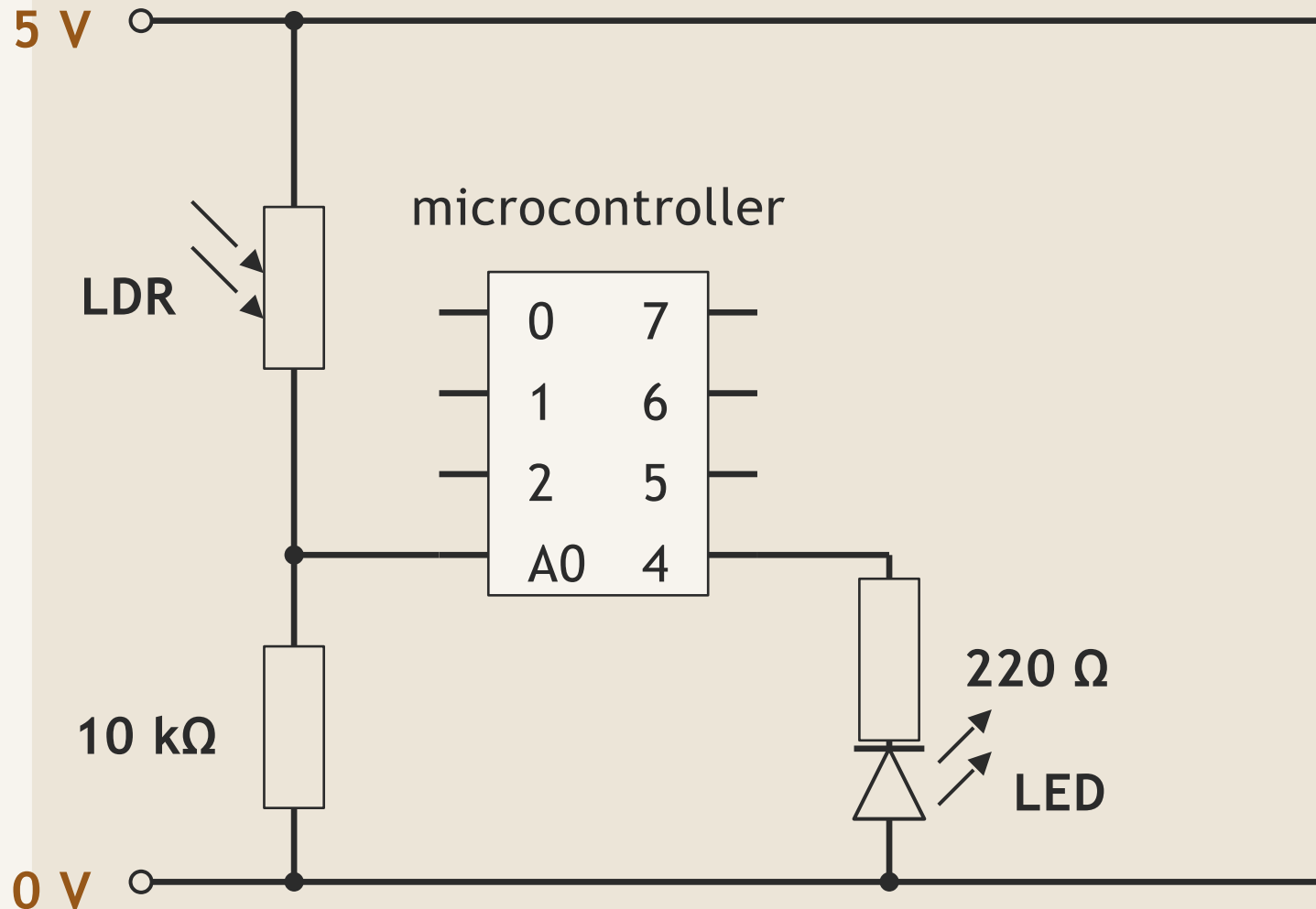
Make it see the dark. Then find what is wrong.

## By the end of this part...



- 1 I can read a sensor with `analogRead()`
- 2 I can count a fixed loop in Arduino C
- 3 I can find and fix faults in a program

## Worked Example 3: The LDR on A0



**analogRead(A0)**

a number from 0 to 1023

**Dark**

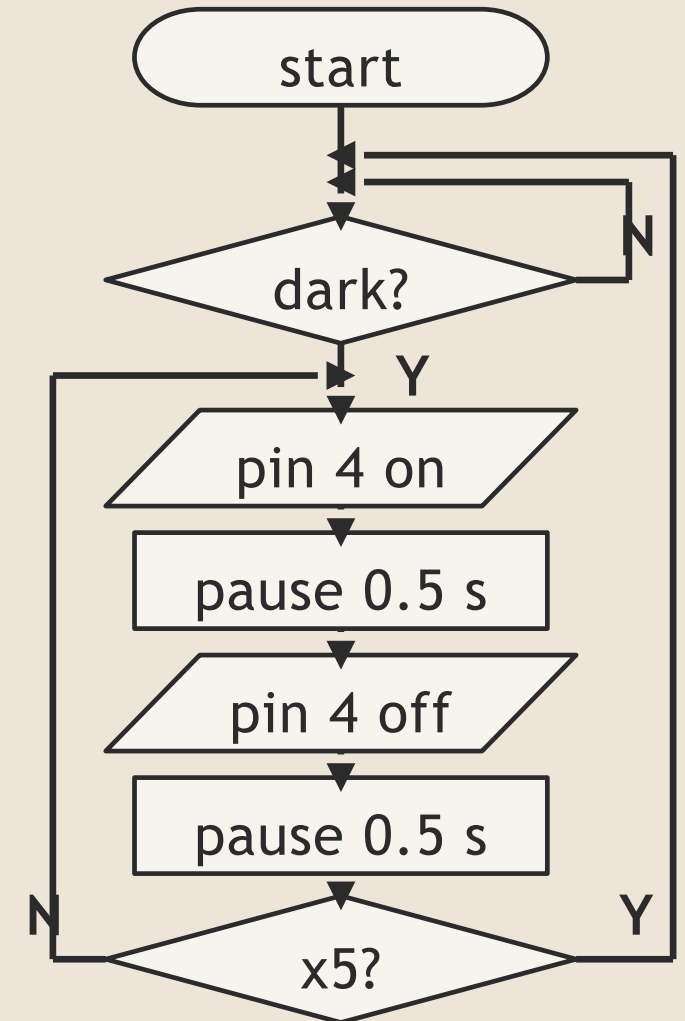
LDR resistance up, A0 down

**light < 300**

means it is dark

## Worked Example 3: Flowchart and Code

```
1  int light = analogRead(A0);  
2  if (light < 300) {  
3    for (int i = 0; i < 5; i++) {  
4      digitalWrite(4, HIGH);  
5      delay(500);  
6      digitalWrite(4, LOW);  
7      delay(500);  
8    }  
9  }
```



# Counting From 0



**for (int i = 0; i < 5; i++)**

**i = 0**

start the count at 0

**i < 5**

keep going while i is under 5

**i++**

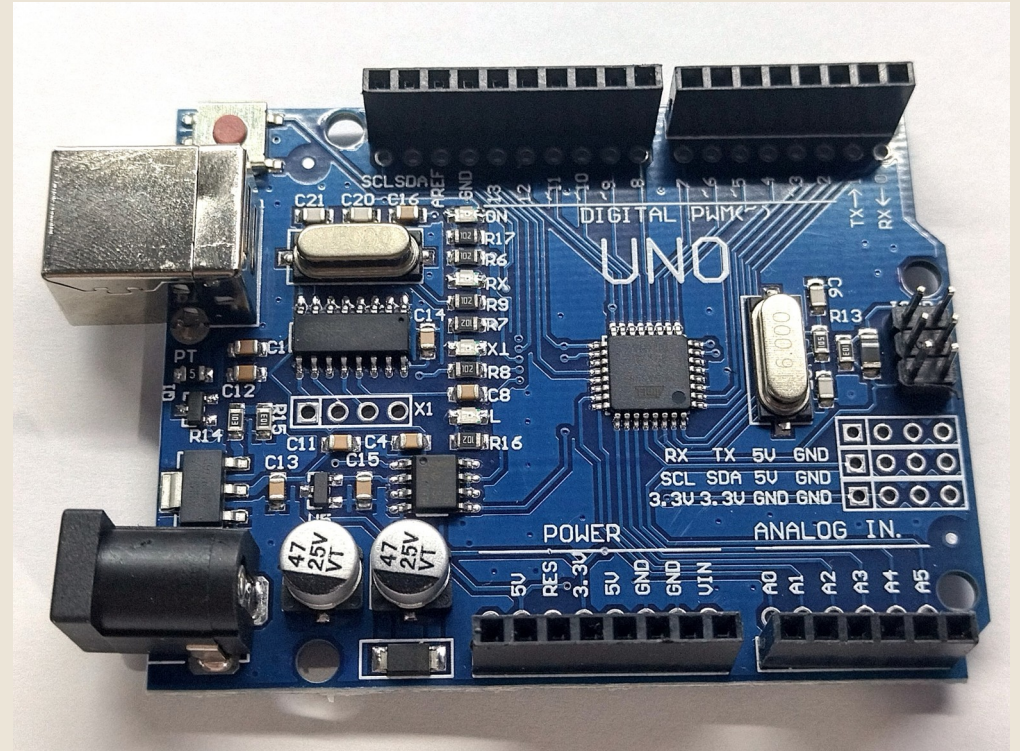
add 1 after each pass

| i | i < 5? | flash |
|---|--------|-------|
| 0 | yes    | 1     |
| 1 | yes    | 2     |
| 2 | yes    | 3     |
| 3 | yes    | 4     |
| 4 | yes    | 5     |
| 5 | no     | stop  |

# What to Collect

Collect one set per pupil.

- 1 LDR
- 2 10 k $\Omega$  resistor (brown, black, orange)
- 3 Push button
- 4 Buzzer
- 5 4 more jumper leads

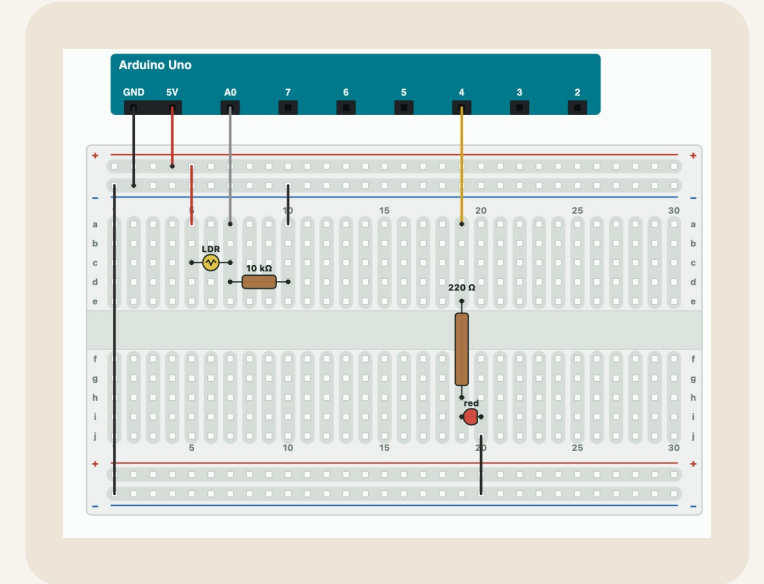


# Build It: Worked Example 3



## Practical sheet 2

Add the LDR and 10 k $\Omega$  (page 18).  
Upload. Cover it: 5 flashes?



**SAFETY** USB power only. Unplug before you move a wire.

15 min



# Try Changing This: Worked Example 3



1 Change 300 to 600. Easier or harder?

Easier — it counts as dark sooner

2  $i < 10$ . How many flashes?

10

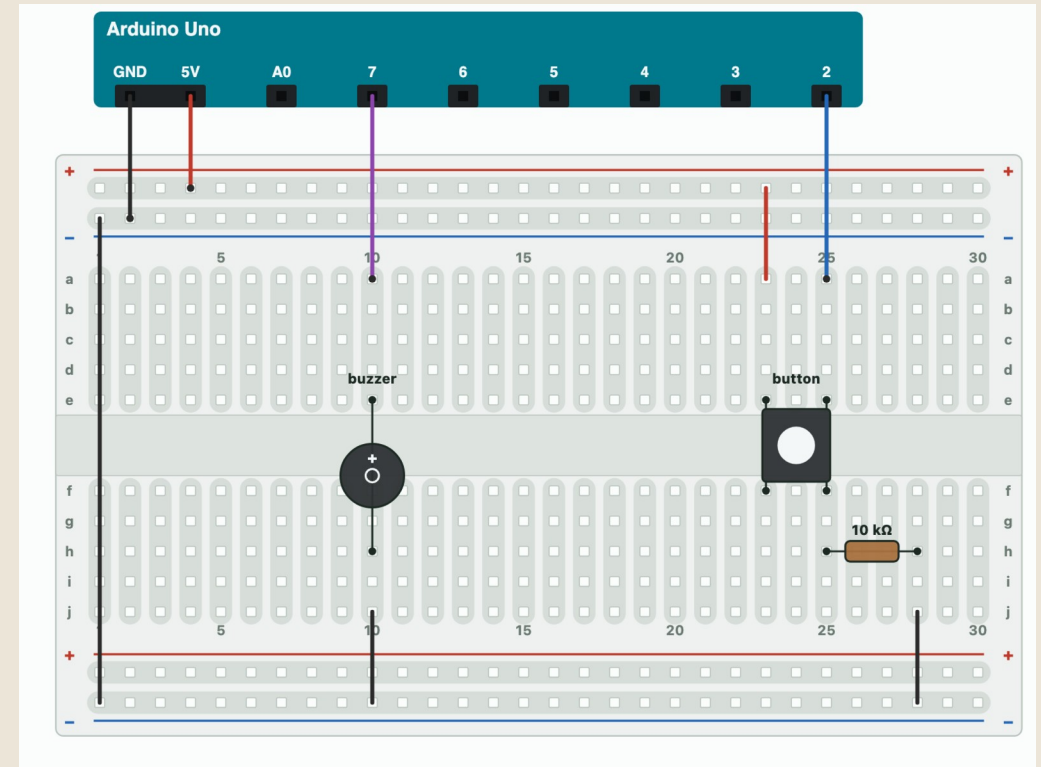
3 Swap  $<$  for  $>$ . When does it go off?

In the light

# Debugging: The Method



- 1 What should it do?
- 2 What does it do?
- 3 Which line?
- 4 Change one thing
- 5 Test it again



# The Faults to Look For



**No pinMode**

the pin is never made an output

**delay(1)**

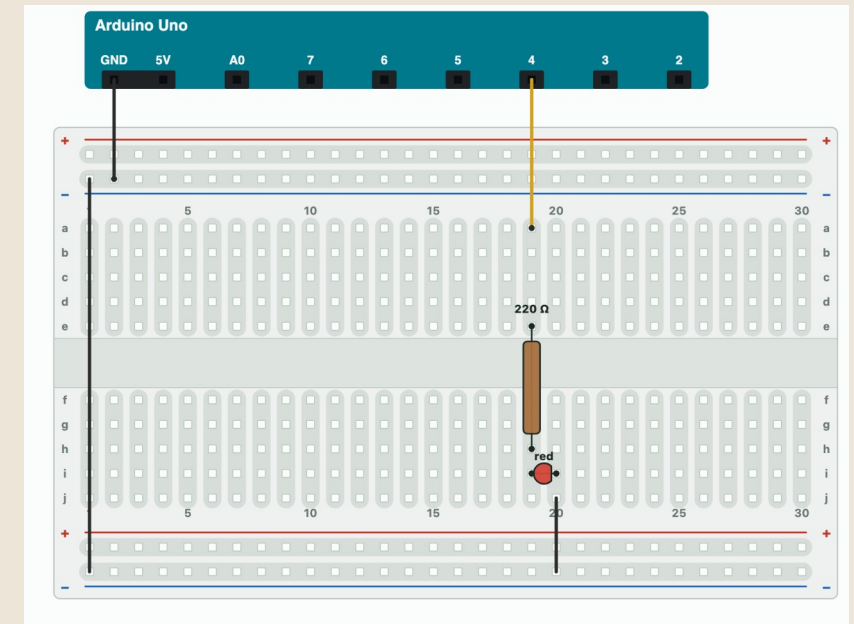
1 ms, not 1 s

**= in an if**

compare with ==

**A missing }**

every { needs its }

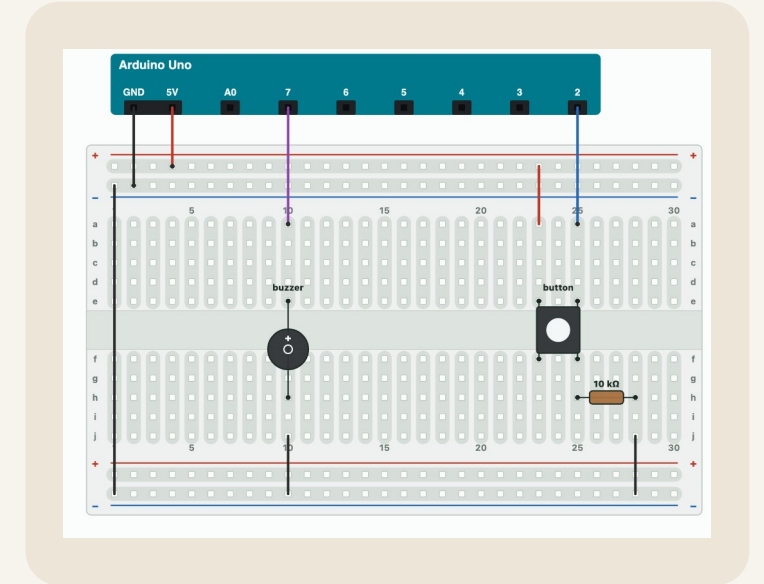


# Find and Fix: Debug Builds 1 and 2



## Practical sheet 2

Build each one (pages 20-21).  
Find 2 faults in each. Fix, re-test.



**SAFETY** USB power only. Unplug before you move a wire.

15 min



# Find and Fix: Debug Build 1



```
1 void setup() {  
2 }  
3 void loop() {  
4   digitalWrite(4, HIGH);  
5   delay(1);  
6   digitalWrite(4, LOW);  
7   delay(1000);  
8 }
```

Should: flash pin 4, 1 s on, 1 s off.

## 1 Line 2

Add pinMode(4, OUTPUT);

## 2 Line 5

delay(1000);

## Find and Fix: Debug Build 2



```
1 void setup() {  
2   pinMode(2, INPUT);  
3   pinMode(7, OUTPUT);  
4 }  
5 void loop() {  
6   if (digitalRead(2) = HIGH) {  
7     digitalWrite(7, HIGH);  
8     delay(2000);  
9     digitalWrite(7, LOW);  
10  }
```

Should: buzz 2 s when the button is pressed.

### 1 Line 6

== HIGH, not = HIGH

### 2 Line 11

Add } to close loop()

# Watch Out: The Exam Never Asks for Code

## THE MISTAKE

- `digitalWrite(4, HIGH);`
- `delay(1000);`
- Code in an exam answer.

## THE CORRECT ANSWER

- pin 4 on
- pause 1 s
- A pin number and a unit.

# Checkpoint



- 1 analogRead(A0) gives a number from...
- 2 for (i = 0; i < 3; i++) runs how often?
- 3 Why change one thing at a time?

0 to 1023

3 times

So you know which  
change fixed it

# Plenary: Part 2



1 In the dark, A0's reading goes...

Down

2 == or = to compare?

==

3 Every { needs...

Its }

4 The fixed loop in the flowchart is...

x5? (a decision)

5 In the exam, never write...

Arduino C

## Look back — by the end of this part...



- 1 I can read a sensor with `analogRead()`
- 2 I can count a fixed loop in Arduino C
- 3 I can find and fix faults in a program

# Daily Review



1 delay(1) waits...

1 ms

2 = or == in an if?

==

3 A0 needs a pinMode line?

No

4 Debugging step 4?

Change one thing

5 The exam's answer is never in...

Arduino C

## PART 3

# Test, Amend, Re-test

It works once. Does it do every job?

## By the end of this part...



- 1 I can write a test for each job a program does
- 2 I can say what happened, not just 'it failed'
- 3 I can plan a change and test it again

# Why Engineers Test

## Job 1

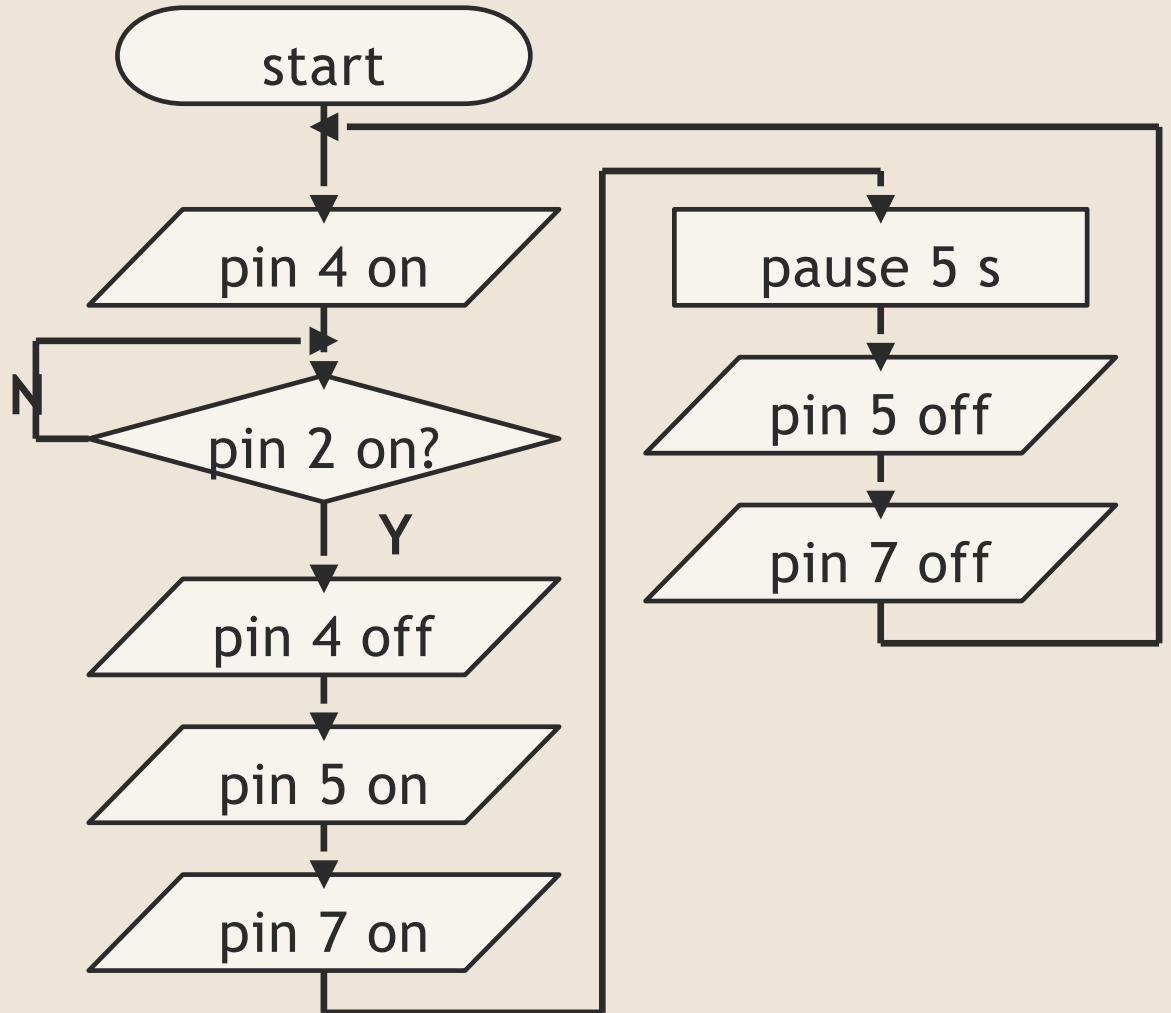
red until the button is pressed

## Job 2

green and beeper for 5 s

## Job 3

back to red, beeper off



# The Test Table



| Test | Should | Happened | Change | Re-test |
|------|--------|----------|--------|---------|
|      |        |          |        |         |
|      |        |          |        |         |
|      |        |          |        |         |

Should = what should happen   •   Happened = what did happen

# Worked Example: The Crossing

| Test     | Should               | Happened             | Change        | Re-test     |
|----------|----------------------|----------------------|---------------|-------------|
| Press    | Red off,<br>green on | Red off,<br>green on | None          |             |
| Wait 5 s | Green off,<br>red on | Green off,<br>red on | None          |             |
| Beeper   | Stops after<br>5 s   | Stays on             | Add pin 7 off | Stops: pass |

# The Fault, in the Sim

Last box

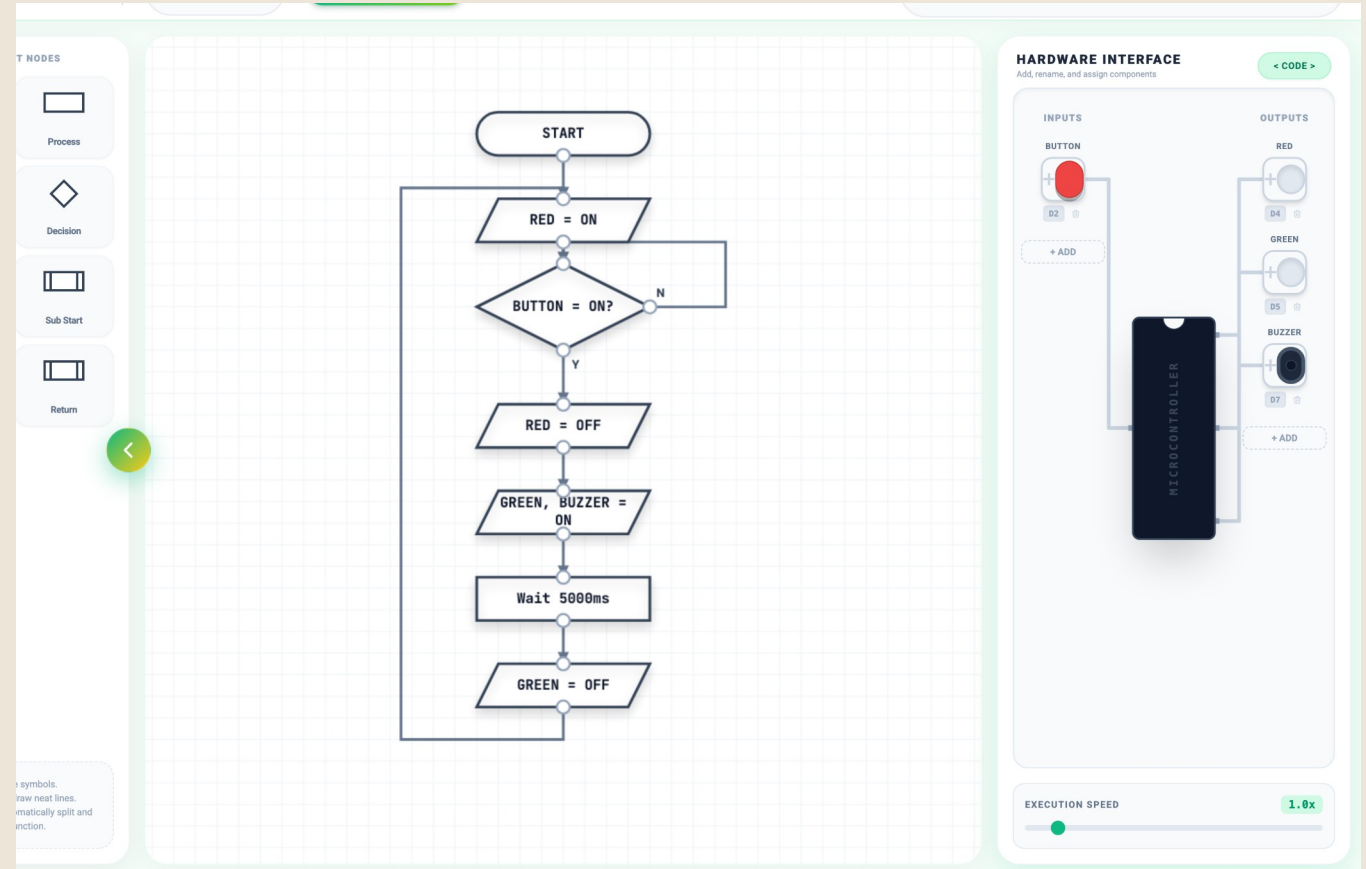
GREEN = OFF

Missing

BUZZER = OFF

So

the beeper never stops



# Say What You Saw

## THE MISTAKE

- It didn't work.
- Fixed it.
- Works now.

## THE CORRECT ANSWER

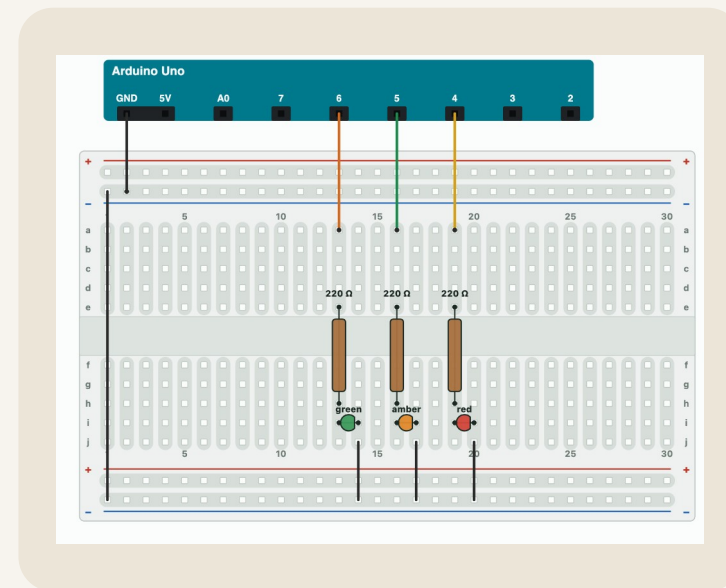
- Beeper stayed on after 5 s.
- Add pin 7 off after pin 5 off.
- Re-test: beeper stops at 5 s.

# Test It: Your Traffic Lights



## Practical sheet 3

Write 3 tests for your traffic lights.  
Run them. Fill in every column.



**SAFETY** USB power only. Unplug before you move a wire.

12 min

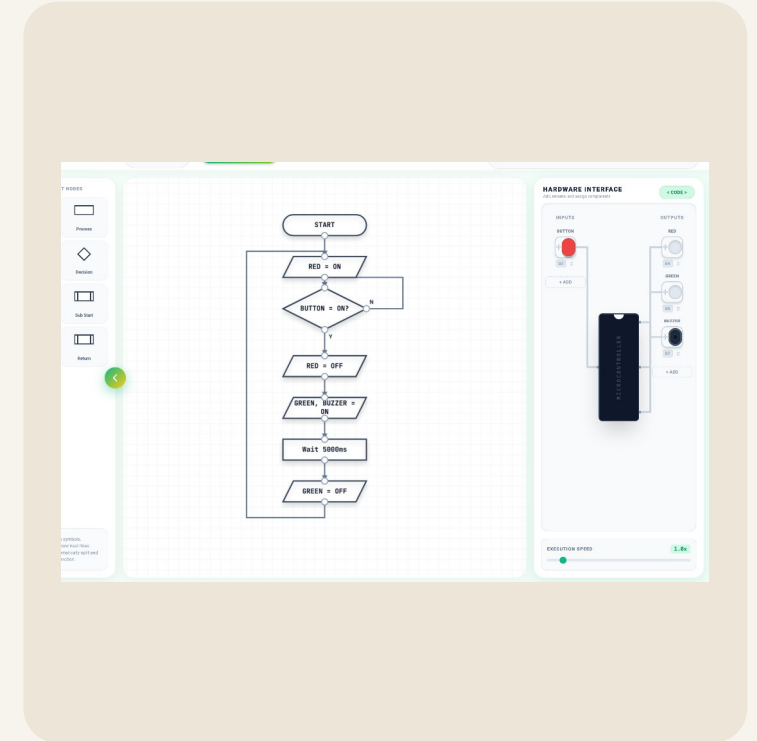


# Test It: The Faulty Crossing



## Practical sheet 3

NoSyntaxSim, then Yenka. Test all 3 jobs.  
Fill the table. Fix it. Re-test.



15 min



# PRACTICE: Programming in Real Life



## Booklet page 23

Q7 and Q8, on your own.  
Then test Q7 on your Uno.

6. Write pseudocode that flashes an LED 4 times, on 0.5 s and off 0.5 s, then stops. (4)

7. Write Arduino C that sounds a buzzer on pin 8 for 3 s when a button on pin 2 is pressed, repeating forever. (4)

8. Convert this pseudocode to Arduino C: loop / high LED / pause 1 second / low LED / pause 1 second / end loop. (4)

23

12 min



# Programming in Real Life: Answers



1 Q7 setup()

```
pinMode(2, INPUT);  
pinMode(8, OUTPUT);
```

2 Q7 loop()

```
if (digitalRead(2) ==  
    HIGH) ...
```

3 Q7 inside the if

```
    pin 8 HIGH,  
    delay(3000), LOW
```

# Programming in Real Life: Answers



4 Q8 setup()

pinMode for the LED's  
pin

5 Q8 loop()

HIGH, delay(1000),  
LOW, delay(1000)

# Plenary: Part 3



- 1 One test for each...
- 2 'It didn't work' is weak. Why?
- 3 After a change you must...
- 4 Which column says what you saw?
- 5 Change how many things at once?

Job the program does

It names no part and no  
result

Re-test

Happened

One

# Extension

EXTRA PRACTICE



1 Add amber to the crossing. Where?

Between green off and  
red on

2 Make the beeper pulse. How?

On, pause, off, pause,  
in a fixed loop

3 The button is never read. Fault?

Its pin is not pin 2 in  
the code

# Look back — by the end of this part...



- 1 I can write a test for each job a program does
- 2 I can say what happened, not just 'it failed'
- 3 I can plan a change and test it again

LOOKING AHEAD

# Booklet 4a: Digital Signals and Logic States

From programs to logic gates.

# Course Specification

## Digital electronic control systems

- using suitable commands (high, low, for...next, if...then, pause, end (or their equivalents)) to design programs to solve simple control problems, involving time delays and continuous and fixed loops

# Credits (1 of 2)

**The big question (An Arduino Uno):** Photo: JrawX, CC0, via Wikimedia Commons

**From Flowchart to Board (Uno photo):** Photo: JrawX, CC0, via Wikimedia Commons

**From Flowchart to Board (NoSyntaxSim):** NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims ([sites.google.com/view/nostrainsim](https://sites.google.com/view/nostrainsim)), free and open to all.

**What to Collect (Uno photo):** Photo: JrawX, CC0, via Wikimedia Commons

**Breadboard and wiring drawings (all slides):** Breadboard and wiring drawings by R Stewart (Stripboard Builder engine).

**Watch Out (course specification):** © Qualifications Scotland (SQA). N5 Engineering Science course specification v2.0, p.9 (question paper).

**The Fault, in the Sim:** NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims ([sites.google.com/view/nostrainsim](https://sites.google.com/view/nostrainsim)), free and open to all.

# Credits (2 of 2)

## **Test It: The Faulty Crossing (NoSyntaxSim):**

NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims

([sites.google.com/view/nostrainsim](https://sites.google.com/view/nostrainsim)), free and open to all.

**PRACTICE: Programming in Real Life (booklet page):** Booklet 4b pages by R Stewart.

**Course Specification:** © Qualifications Scotland (SQA). Course spec v2.0, p.5